How to Run: Walkthrough

NLP Curriculum Bridge · Class 6 Science topic preprocessing

Pages 2–9: setup, run and check, step by step, with the terminal output you should see.
Pages 10–12: the final output (topics and chunks per book, sample topics, one full chunk record).

You only need to run it to regenerate the output. **processed/chunks.jsonl** is already in the repo and ready for the embedding step (use section = content).

STEP 1 OF 8

Get the latest code

If you already have the repo, pull. If not, clone it first: git clone https://github.com/Harshitha-arch/NLP-Curriculum-Bridge-Migrant-Kids-.git

Command

```
git pull
```

What you should see

```
$ git pull
Already up to date.
```

STEP 2 OF 8

Install uv (one time)

uv installs the right Python packages for the project. Install it, then check it works.

Command

```
pip install uv
uv --version
```

What you should see

```
$ uv --version
uv 0.12.17 (635500036 2026-09-18 x86_64-pc-windows-msvc)
```

STEP 3 OF 8

Install Tesseract OCR (one time)

Needed for a few figure pages with no text layer. On Windows it isn't added to PATH, but the pipeline finds it in the default folder automatically, so check it with the full path.

Command

```
Windows: winget install UB-Mannheim.TesseractOCR
Mac: brew install tesseract
```

What you should see

```
$ & "C:\Program Files\Tesseract-OCR\tesseract.exe" --version
tesseract v5.4.0.20240606
 leptonica-1.84.1
```

STEP 4 OF 8

Install the project's dependencies

Run this inside the repo folder. The first time, it downloads the packages, so you'll see a longer list.

Command

```
uv sync
```

What you should see

```
$ uv sync
Resolved 23 packages in 1ms
Checked 22 packages in 2ms
```

STEP 5 OF 8

Run the preprocessing

Reads every PDF and writes the chunks and topics to processed/. The first run takes a few minutes; after that about a minute.

Command

```
uv run python -m preprocessing.run
```

What you should see

```
$ uv run python -m preprocessing.run
data\NCERT\fecu101.pdf: 246 lines, 0 OCR pages
data\NCERT\fecu102.pdf: 1153 lines, 0 OCR pages
data\NCERT\fecu103.pdf: 1187 lines, 1 OCR pages
data\NCERT\fecu104.pdf: 779 lines, 0 OCR pages
data\NCERT\fecu105.pdf: 775 lines, 0 OCR pages
data\NCERT\fecu106.pdf: 908 lines, 1 OCR pages
data\NCERT\fecu107.pdf: 923 lines, 1 OCR pages
data\NCERT\fecu108.pdf: 914 lines, 0 OCR pages
data\NCERT\fecu109.pdf: 976 lines, 1 OCR pages
data\NCERT\fecu110.pdf: 1068 lines, 1 OCR pages
data\NCERT\fecu111.pdf: 947 lines, 0 OCR pages
data\NCERT\fecu112.pdf: 1008 lines, 0 OCR pages
data\SCERT\Term_1.pdf: 6035 lines, 0 OCR pages
data\SCERT\Term_2.pdf: 5702 lines, 0 OCR pages
data\SCERT\Term_3.pdf: 4978 lines, 0 OCR pages
kept 294 topics as 705 chunks, dropped 6 -> processed
```

STEP 6 OF 8

Optional: check the output against the PDFs

Goes through every line of text in every PDF and checks it is in the output or was removed for a known reason (page numbers, footers, cover/glossary pages, verses). The last column must be 0.

Command

```
uv run python -m preprocessing.verify
```

What you should see

```
$ uv run python -m preprocessing.verify
book                lines     found   outside     verse furniture   dropped   heading   unexpl.
NCERT/fecu101.pdf     246       217         0         0        21         0         0         0
NCERT/fecu102.pdf    1153       904         0         5       116         0         5         0
NCERT/fecu103.pdf    1187       977         0         4        91         0        11         0
NCERT/fecu104.pdf     779       556         0         0       128         0         5         0
NCERT/fecu105.pdf     775       614         0         0        82         0         7         0
NCERT/fecu106.pdf     908       708         0         4        85         0         8         0
NCERT/fecu107.pdf     923       693         0         0       106         0         2         0
NCERT/fecu108.pdf     914       724         0         0        72         0         7         0
NCERT/fecu109.pdf     976       732         0         4       142         0         0         0
NCERT/fecu110.pdf    1068       860         0         0       108         0         7         0
NCERT/fecu111.pdf     947       768         0         0        97         0         8         0
NCERT/fecu112.pdf    1008       826         0         0       107         0         5         0
SCERT/Term_1.pdf     6035      4866       332         0       630         0        18         0
SCERT/Term_2.pdf     5702      4351       317         0       689         0        51         0
SCERT/Term_3.pdf     4978      3817       216         0       730         2        54         0
TOTAL               27599     21613       865        17      3204         2       188         0
columns: outside = cover/contents/glossary/credits pages, furniture = page numbers,
footers, labels; unexpl. = missing for no known reason (should be 0)
missing lines written to processed\verification.csv
```

STEP 7 OF 8

Optional: run the tests

Checks the cleaning, splitting and chunking rules. All 43 should pass.

Command

```
uv run pytest
```

What you should see

```
$ uv run pytest
============================= test session starts =============================
platform win32 -- Python 3.12.14, pytest-9.1.1, pluggy-1.7.0
rootdir: C:\Users\yuvra\dev\NLP-Curriculum-Bridge
configfile: pyproject.toml
testpaths: tests
plugins: hypothesis-6.170.5
collected 43 items

tests\test_clean.py ...........................                          [ 67%]
tests\test_run.py ..                                                     [ 72%]
tests\test_segment.py ............                                       [100%]

============================= 43 passed in 6.55s ==============================
```

STEP 8 OF 8

Check the output

chunks.jsonl is the embedding input (use section = content). topics.jsonl has each topic's full text; verification.csv comes from the optional verify step.

Command

```
ls processed
```

What you should see

```
$ ls processed
chunks.jsonl
dropped_topics.jsonl
summary.csv
topics.jsonl
verification.csv
```

Preprocessing output: topics per book

294 topics kept as 705 chunks (545 content chunks), 6 dropped (empty or too short).

Book	Topics found	Topics kept	Chunks
NCERT/fecu101.pdf	1	1	8
NCERT/fecu102.pdf	9	9	33
NCERT/fecu103.pdf	15	15	35
NCERT/fecu104.pdf	10	10	21
NCERT/fecu105.pdf	11	11	24
NCERT/fecu106.pdf	12	12	25
NCERT/fecu107.pdf	7	7	23
NCERT/fecu108.pdf	11	11	29
NCERT/fecu109.pdf	4	4	24
NCERT/fecu110.pdf	10	10	34
NCERT/fecu111.pdf	12	12	30
NCERT/fecu112.pdf	15	15	32
SCERT/Term_1.pdf	43	38	132
SCERT/Term_2.pdf	74	74	137
SCERT/Term_3.pdf	66	65	118
Total	**300**	**294**	**705**

Sample topics (text shortened)

Each topic: title, where it comes from, and the clean text that gets embedded.

Magnetic and Non-magnetic Materials

NCERT · Exploring Magnets · content · pages 2-4

Magnetic and Non-magnetic Materials. Activity 4.1: Let us explore Collect a few objects made of different materials and also a magnet. Predict which of the objects will stick to the magnet. Write your prediction in Table 4.1. Now hold a magnet in your hand and bring it near the objects one by one (Fig. 4.3). Observe which of the...

Balanced Diet

NCERT · Mindful Eating: A Path to a Healthy Body · content · pages 17-19

Balanced Diet. Are nutritional requirements the same for everyone? Do you and your grandparents need the same type or the same amount of nutrients? Requirements of the type and amount of nutrients in a diet may vary according to age, gender physical activity, health status, lifestyle, and so on. Activity 3.8: Let us find out You...

Keywords and Summary

NCERT · Measurement of Length and Motion · summary · pages 18-18

Keywords and Summary. Keywords Classify Justification Identify Explore Investigate Observe Linear motion Length kilometre Distance Circular motion centimetre Measurement SI Unit of Length Reference point Oscillatory motion Motion millimetre metre The International System of Units (SI units) has been adopted by countries as stand...

Mass

SCERT · Measurements · content · pages 11-12

Mass is the measure of the amount of matter in an object. The SI unit of mass is kilogram. It is represented by 'kg'. Weight is the gravitational pull experienced by matter. The weight is directly proportional to the mass on the Earth's surface. Hold a sheet of paper in one hand and a book in other hand. Which hand feels the hea...

Expansion in solids

SCERT · Heat · content · pages 13-15

Expansion in solids. Sam is trying to open a tight jar, but he cannot open it. He asks his uncle to help. His uncle says that pour some hot water on the lid of the jar. Sam does so and tries to open it now. Wow! The jar is opened easily! Do you have such experience? How do you open a tightly closed cap of the THERMAL EQUILIBRIUM...

One record in chunks.jsonl

Text shortened for the screenshot.


```
{
  "chunk_id": "SCERT_c6_Term_1_03_2",
  "topic_id": "SCERT_c6_Term_1_03",
  "board": "SCERT",
  "grade": 6,
  "term": 1,
  "source_file": "SCERT/Term_1.pdf",
  "chapter": "Measurements",
  "topic_title": "Mass",
  "section": "content",
  "page_start": 11,
  "page_end": 12,
  "ocr": false,
  "lang": "en",
  "chars": 1015,
  "chunk_index": 2,
  "n_chunks": 2,
  "text": "Mass. 1000 milligram = 1 gram 1000 gram = 1 kilogram 1000 kilogram = 1 tonne Measuring the length of a curved line using a divider. Draw a curved line AB on a piece of paper. Separate the legs of the divider by 0.5 cm or 1 cm using a ruler. Place it on the curved line starting from one end. Mark the position of the other end. Move it along the line again and again cutting the line into number of segments of equal len ..."
}
```